Consumer Shopping Behavior Analysis

Leveraging Data to Optimize Retail Strategy and Customer Engagement

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Executive Summary

End-to-End Analysis

Identified purchasing trends across demographics and channels.

Drive Loyalty & Sales

Goal: long-term loyalty and increased sales.

Actionable Intelligence

Python, SQL, Power BI transform raw data into insights.

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Problem Statement

Understanding Shifts

Management sought drivers behind repeat purchases and decision-making.

> "How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

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Dataset Summary

3,900 Rows

Extensive customer and purchase data.

18 Columns

Demographics, purchase details, shopping behavior.

Key Features

Age, Gender, Location, Item Purchased, Amount, Reviews.

Missing Data

37 values in Review Rating column.

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Exploratory Data Analysis (Python)

Data Preparation

- Data Loading (pandas)
- Initial Exploration (df.info(), .describe())
- Missing Data Handling (median imputation)
- Column Standardization (snake_case)

Feature Engineering

- Age groups, purchase frequency days.
- Dropped redundant promo_code_used.
- Loaded cleaned data to PostgreSQL.

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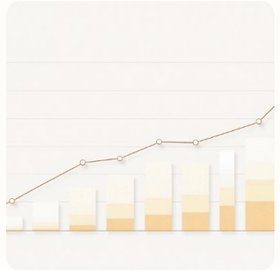
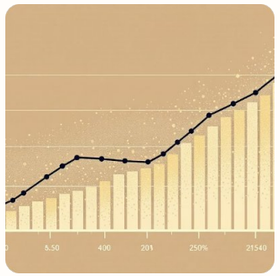
Data Analysis Using SQL



Gender-Based Revenue

Spending differences between male and female customers.



Discount Impact

Customers using discounts with above-average spend.

Product Rating

Top 5 products with highest average review ratings.

Shipping Type

Average purchase amounts for Standard vs. Express shipping.

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SQL Analysis: Key Insights

Subscription Revenue

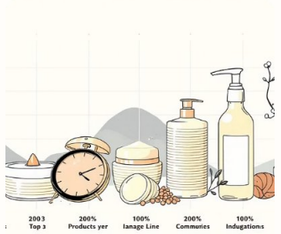
Subscribed customers' average spend and total revenue.

Discount Utilization

Top 5 products with highest percentage of discounted purchases.

Customer Segmentation

New, Returning, and Loyal categories based on purchase history.



Category-Wise Ranking

Top 3 most purchased products within each category.

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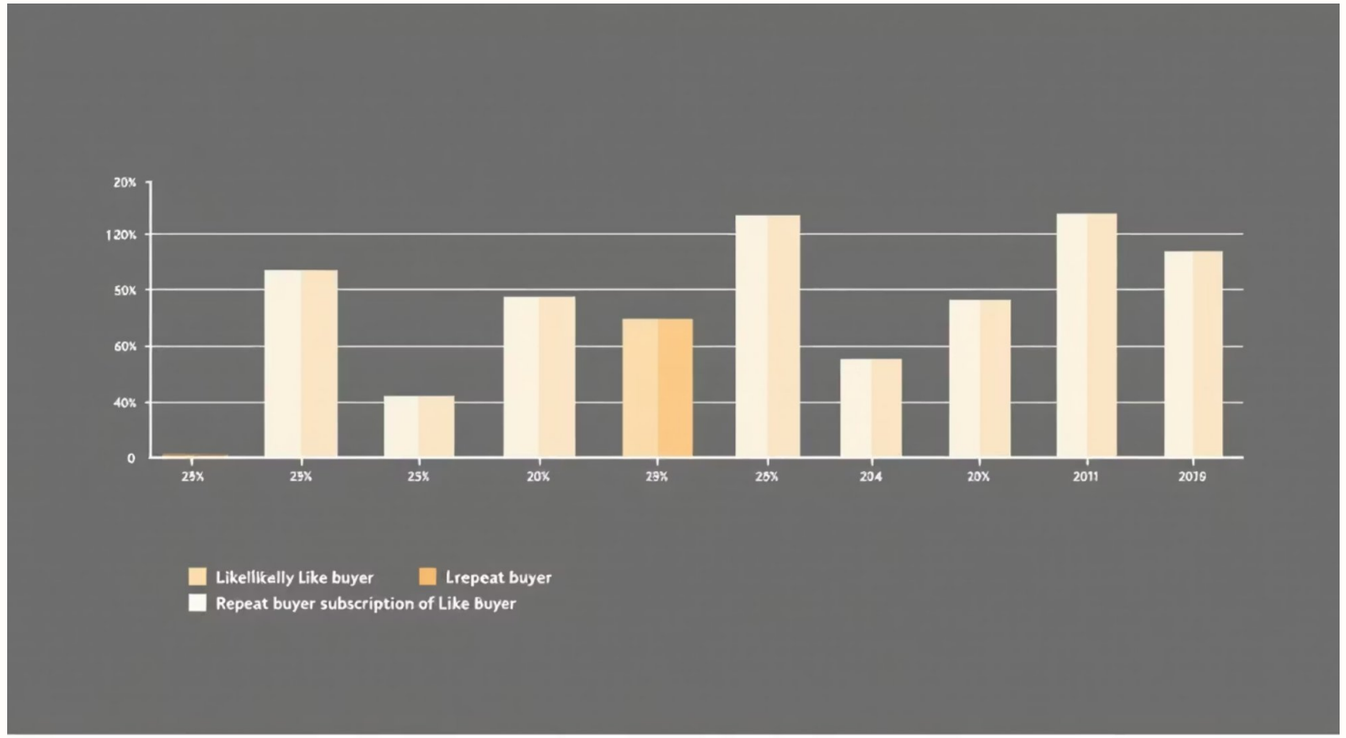
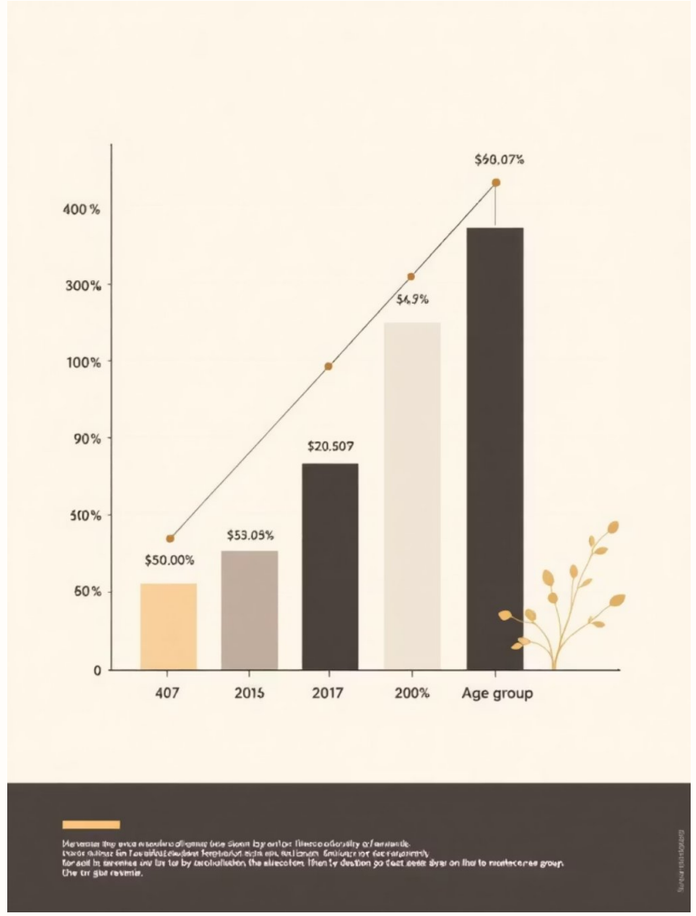
SQL Analysis: Deeper Dive



Repeat Buyer Subscription

Are repeat buyers more likely to subscribe?



Age Group Revenue

Revenue contribution across different customer age groups.

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Interactive Dashboard in Power BI

Visualizing insights for stakeholders.

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Business Recommendations

1 Personalized Marketing

Tailor discount offers based on demographics and payment history.

2 Omnichannel Synergy

Bridge online reviews and offline shopping to boost confidence.

3 Seasonal Inventory

Align stock levels with peak periods to maximize sales.

Deliverables

- Python Scripts, SQL Queries, Power BI Files, GitHub Repository.

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Customer Behavior Report

Subscription status

No | Yes

Gender

Female | Male

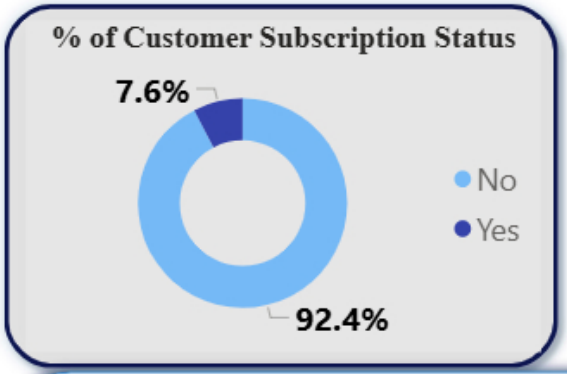
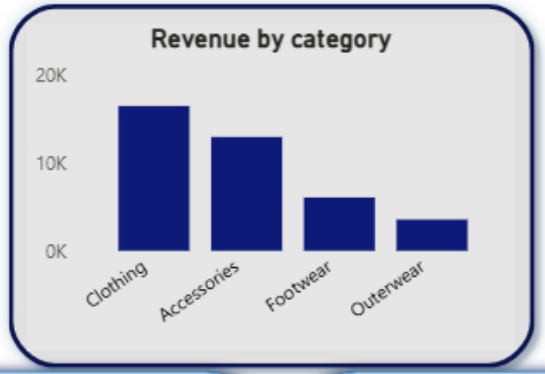
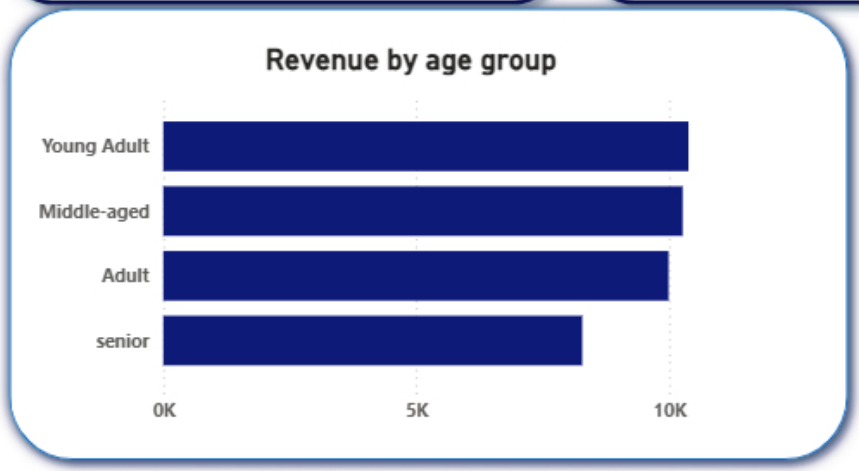
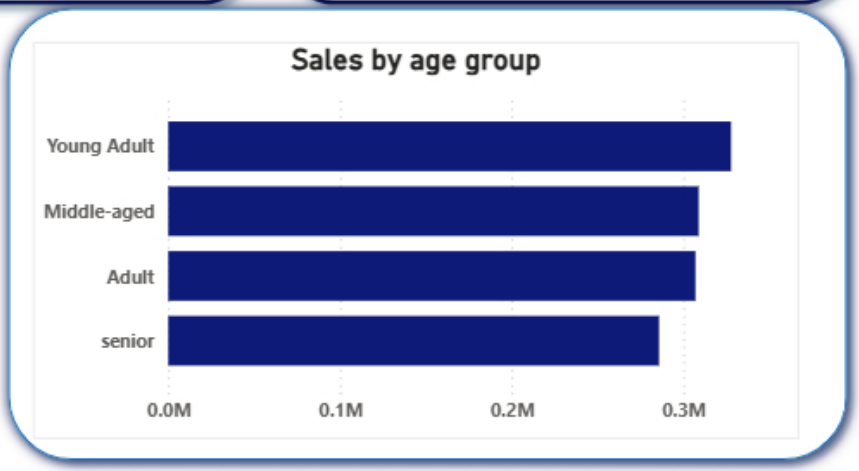
Category

- Accessories
- Clothing
- Footwear
- Outerwear

Shipping type

- 2-Day Shipping
- Express
- Free Shipping
- Next Day Air
- Standard
- Store Pickup

$59.89
Average Purchase Amount

650
Number of Customer

3.71
Average Review Rating